

DRM: Directional Relational Manifolds

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Abstract

We introduce Directional Relational Manifolds (DRM), a class of geometrical structures in which dimensionality is not fixed but emerges dynamically from relational directional fields. Unlike classical manifolds based on orthogonal tangent spaces, DRMs define dimensions as active directions whose structure may vary locally and globally. We develop the formal foundations of DRM, define its metric, connection, curvature, and dynamics, and show that stable DRMs naturally converge to toroidal topology. DRM extends information geometry [Amari and Nagaoka(2000)] from the fixed-dimension setting to variable-dimension structures, reducing to the Fisher–Rao framework as a special case. This framework provides a unified mathematical language for adaptive physical, cognitive, and informational systems.

1 Introduction

Modern geometry assumes that space is composed of fixed-dimensional manifolds locally modeled by $\mathbb{R}^n$. While successful in classical physics, this assumption becomes inadequate for systems whose degrees of freedom evolve dynamically, such as turbulent fluids, quantum fields, neural systems, and adaptive artificial intelligence.

We propose a new geometric paradigm in which dimensionality is not primitive but emergent. In Directional Relational Manifolds, dimensions are defined as locally active directions of interaction, allowing infinite and variable-dimensional structures.

2 Motivation

Classical manifolds rely on three restrictive assumptions:

1. Fixed global dimensionality
2. Orthogonal tangent bases
3. Static geometric structure

However, real systems exhibit:

- Locally varying degrees of freedom
- Interacting non-orthogonal modes
- Self-modifying state spaces

These observations motivate a geometry in which space itself adapts.

2.1 Relation to Information Geometry

Information geometry [Amari and Nagaoka(2000)] endows the space of probability distributions with a Riemannian structure via the Fisher Information Matrix:

$$g_{ij}(\theta) = \mathbb{E} \left[\frac{\partial \log p}{\partial \theta_i} \cdot \frac{\partial \log p}{\partial \theta_j} \right]. \quad (1)$$

This framework is intrinsic (invariant under reparametrisation by Chentsov’s theorem [Chentsov(1982)]) and unifies statistical estimation, exponential families, and divergence geometry under a single formalism [Nielsen(2020)].

DRM differs from information geometry in one structural dimension: Amari’s framework operates on manifolds of *fixed* dimension n , where the parameter space $\Theta \subseteq \mathbb{R}^n$ has constant rank. DRM generalises this to allow $\dim_{\mathcal{D}}(p)$ to vary pointwise—modelling systems whose active degrees of freedom emerge, collapse, or interact non-orthogonally as a function of state.

This generalisation is motivated by cognitive and physical systems that cannot be adequately described by a fixed-rank parameter space: a reasoning system mid-collapse has fewer effective degrees of freedom than the same system in a healthy state, and this reduction is itself a dynamically observable quantity, not a fixed property of the model class.

When $\dim_{\mathcal{D}}(p) = n$ (constant) and g_p is the Fisher metric on a statistical model, DRM reduces to information geometry (Corollary 10.1).

3 Directional Relational Manifolds

Definition 3.1 (Directional Set). *Let M be a topological space. For each point $p \in M$, define the directional set*

$$\mathcal{D}(p) = \{v \mid v \text{ is an active direction at } p\}. \quad (2)$$

Elements of $\mathcal{D}(p)$ represent relational degrees of freedom.

Definition 3.2 (Dimensionality). *The local dimensionality at p is*

$$\dim_{\mathcal{D}}(p) = |\mathcal{D}(p)|. \quad (3)$$

Dimensionality may vary across M .

Definition 3.3 (Directional Relational Manifold). *A Directional Relational Manifold is a pair $(M, \mathcal{D})$ such that:*

1. *M is a topological space*
2. *$\mathcal{D}$ assigns to each $p \in M$ a measurable directional set*
3. *$\mathcal{D}(p)$ varies continuously in the Hausdorff sense*

4 Relational Metric Structure

Definition 4.1 (Directional Relational Metric). *For each $p \in M$, define*

$$g_p : \mathcal{D}(p) \times \mathcal{D}(p) \rightarrow \mathbb{R} \quad (4)$$

such that:

1. *$g_p(v, v) \geq 0$*
2. *$g_p(v, w) = g_p(w, v)$*
3. *$g_p(v, w)$ measures interaction strength, not orthogonality*

Thus geometry measures relational coupling, not spatial length alone.

Remark 4.1. *When $\mathcal{D}(p)$ is a fixed tangent space $T_p M$ and g_p is the Fisher Information Matrix, the relational metric reduces to the Fisher–Rao metric of information geometry [Amari and Nagaoka(2000)]. The relational interpretation generalises this by allowing g_p to encode interaction strength between non-orthogonal directions of arbitrary cardinality.*

5 Directional Dynamics

The DRM induces a dynamical system on the configuration space $\mathcal{P}$ of directional fields:

$$\dot{x} = F(x), \quad x \in \mathcal{P}. \quad (5)$$

We require the following structural properties:

1. **Boundedness.** The orbit $\{F^t(x)\}_{t \geq 0}$ is contained in a compact subset of $\mathcal{P}$.
2. **Recurrence.** For every $x \in K$ and every open neighbourhood $U \ni x$, the orbit of x returns to U infinitely often.
3. **Structural stability.** Small C^1 -perturbations of F produce topologically conjugate flows on K .

These properties are posited as *design constraints* rather than proven properties of all DRM instantiations. A given implementation may or may not satisfy them depending on hyperparameters and input characteristics. They define physically and cognitively realistic systems and are sufficient conditions for the convergence theorem below.

6 Directional Connection and Curvature

Definition 6.1 (Directional Connection). *Define*

$$\nabla_v^{\mathcal{D}} w \quad (6)$$

as the variation of direction w along v , dependent on the active configuration $\mathcal{D}(p)$.

Definition 6.2 (Directional Curvature).

$$R^{\mathcal{D}}(u, v)w = \nabla_u^{\mathcal{D}} \nabla_v^{\mathcal{D}} w - \nabla_v^{\mathcal{D}} \nabla_u^{\mathcal{D}} w - \nabla_{[u, v]}^{\mathcal{D}} w. \quad (7)$$

This curvature measures deformation of the dimensional structure itself.

7 Geodesics in DRM

Definition 7.1 (Directional Curves). *A curve $\gamma(t)$ is admissible if*

$$\dot{\gamma}(t) \in \text{span}(\mathcal{D}(\gamma(t))). \quad (8)$$

Definition 7.2 (Relational Geodesics). *Let $\Gamma(p, q)$ be the set of admissible curves minimising*

$$\int g_{\gamma(t)}(\dot{\gamma}, \dot{\gamma}) dt. \quad (9)$$

In general, $|\Gamma(p, q)| > 1$. Path uniqueness is not fundamental.

8 Toroidal Convergence

Lemma 8.1. *If DRM dynamics is bounded and recurrent, its ω -limit set is compact and invariant.*

Theorem 8.1 (Toroidal Convergence). *Let $(M, \mathcal{D})$ be a stable DRM satisfying boundedness, recurrence, and structural stability. Then its global attractor satisfies*

$$\mathcal{A}_{\text{DRM}} \cong \mathbb{T}^n. \quad (10)$$

Proof. Bounded recurrent flows generate compact invariant sets. Structural stability implies persistence of quasi-periodic modes. Classical results in dynamical systems [Katok and Hasselblatt(1995)] show that compact invariant sets generated by n independent oscillatory modes are diffeomorphic to the n -torus $\mathbb{T}^n$. $\square$

Remark 8.1 (Applicability conditions). *The three structural properties—boundedness, recurrence, and structural stability—are sufficient conditions for the convergence result. Whether a given implementation of the DRM satisfies them must be verified empirically for each deployment. In particular, structural stability (the exclusion of chaotic dynamics in the directional update rule) is a strong assumption that may not hold for all parameter regimes or input distributions. The theorem establishes what happens if these conditions hold; it does not claim that all DRM instantiations necessarily satisfy them.*

Thus, stable adaptive geometries close topologically.

9 Physical and Cognitive Interpretation

DRM provides a unified geometry for:

- Turbulent fluids
 - Quantum field configuration spaces
 - Neural and cognitive state spaces
 - Adaptive artificial intelligence
- All exhibit emergent, self-modifying dimensional structure.

10 Reduction to Classical Geometry

Theorem 10.1 (Reduction to Riemannian Geometry). *If $\mathcal{D}(p)$ is finite, constant, and orthonormal for all p , DRM reduces to a Riemannian manifold.*

Proof. Constant cardinality $|\mathcal{D}(p)| = n$ for all p recovers a fixed tangent bundle. Orthonormality of $\mathcal{D}(p)$ yields a standard inner product on each tangent space. The directional connection reduces to the Levi-Civita connection, and relational geodesics reduce to Riemannian geodesics by uniqueness. $\square$

Corollary 10.1 (Reduction to Information Geometry). *Let $(M, \mathcal{D}, g)$ be a DRM and let $\mathcal{S} = \{p(x; \theta) \mid \theta \in \Theta \subseteq \mathbb{R}^n\}$ be a parametric statistical model. If:*

1. $\dim_{\mathcal{D}}(p) = n$ for all $p \in M$ (constant dimension);
2. $\mathcal{D}(p)$ spans the tangent space $T_{\theta}\mathcal{S}$ (full span); and
3. $g_p(e_i, e_j) = \mathbb{E}_{p(x; \theta)}[\partial_i \log p \cdot \partial_j \log p]$ (g_p is the Fisher Information Matrix);

then $(M, \mathcal{D}, g)$ is isomorphic to the statistical manifold $(\mathcal{S}, g^F)$ equipped with the Fisher–Rao metric.

Proof. Condition (i) yields constant rank (Theorem 10.1). Conditions (ii)–(iii) identify $g_p|_{\mathcal{D}(p)}$ with the Fisher Information Matrix $g_{ij}^F(\theta)$, which is the canonical Riemannian metric on $\mathcal{S}$ by Chentsov’s theorem [Chentsov(1982)]. The directional connection reduces to the $\alpha = 0$ Levi-Civita connection of information geometry, and relational geodesics recover Fisher–Rao geodesics. $\square$

Corollary 10.1 establishes that information geometry is a special case of DRM where the three conditions (constant dimension, full span, Fisher metric) hold simultaneously. DRM admits strictly richer structure when any of the three conditions is relaxed.

Thus DRM strictly generalises both classical Riemannian geometry and information geometry.

11 Conclusion

We introduced Directional Relational Manifolds, a new geometrical foundation in which dimensionality is emergent, relational, and dynamic. DRM unifies geometry and dynamics, showing that stable adaptive spaces possess intrinsic toroidal topology. By establishing formal reduction results to both Riemannian geometry (Theorem 10.1) and information geometry (Corollary 10.1), we position DRM as a proper generalisation of existing frameworks while maintaining full compatibility with established results. This framework opens a new research programme for mathematics, physics, and cognitive science.

References

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